

Data Exploration

This document details some of the introductory data exploration done to understand how many people are using the product. There is a paired notebook that goes along with this document to show the code.

Basic Information

In the original data set of app reviews, there are 6 columns: `appID`, `reviewerName`, `reviewText`, `reviewerRating`, `reviewDate`, and `textAnalytics`. There are also 111143 rows of data

`appID`

There are 492 apps that have been reviewed in this data set. The range of reviews per app is quite large. The app with the most reviews is `app-2078` with 4,213 reviews, and there are 35 apps with only 1 review (the lowest amount of reviews possible)

`reviewerName`

With an initial query to find the unique count of reviewers, there was a count of 79,372 unique users submitting reviews. However, when looking at a value count, about 20,000 of those users were listed as "A Google User". These indicate guest user reviews, and so they should be counted as individual users to get an accurate account of how many people would be submitting reviews. So with this in mind, the total number of unique Users submitting reviews is around 100,000. Further, the user with the highest amount of reviews is John Smith with 26, followed by 6 users with at least 10 reviews.

`reviewText`

There are 100,936 unique written reviews. Most of the reviews, 98,830, are only seen once in the entire dataset, typically indicating these are at least a sentence long or contain a unique variable of text. The rest are simple and easily repeatable strings of text, such as one word responses such as "Good", "Ok", or "Great!". Some of these simple ones are further separated by capitalization inconsistencies, so "Nice" and "nice" are considered different variables. Would likely want to capitalize all values to remove these discrepancies.

`reviewerRating`

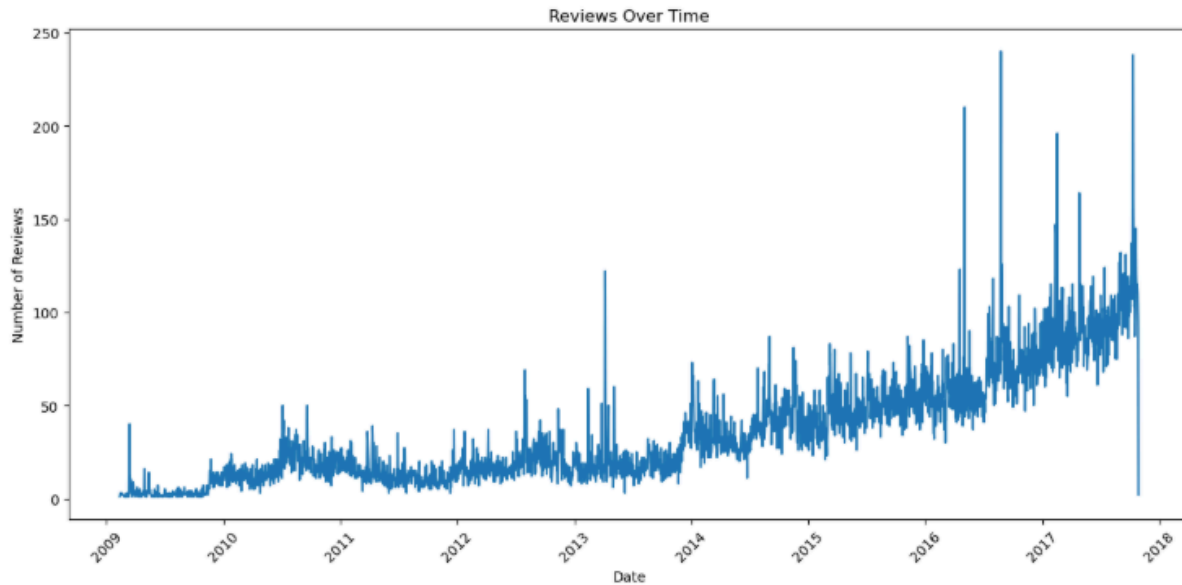
There are 6 different levels of this variable, pictured below:

0.0	71
0.4	5244
0.6	9160
0.2	10296
0.8	18719
1.0	67653

There is a big skew towards higher ratings. The reviews of 0 is something to take note of.

reviewDate

Originally, this date column had multiple different formattings for the data points. To fix this inconsistency, i converted everything into a yyyy/mm/dd format. The dates of reviews are sporadic, but the data starts from 2009/02/11 and ends at 2017/10/28. After plotting the counts of reviews by date, there is a steady increasing trend of reviews as the years pass.



textAnalytics

Im not fully sure what the meaning of this column is, but most of the values in this column are null (110,656). The rest of the 487 rows that have values in them, they seem to be on a range from 0.0 to 100.0. Looking at the other data in the row does not give good indication of what it means.

Key Take Aways

There about 100,000 individual end-users for this mobile app store contribution around 111,000 reviews over the course of 8 years. The amount of reviews is shown to be increasing over time, so we should expect to be able to handle at least 100,000 more reviews in the coming years. The model should account for the computational power of this. There is also about 500 individual apps being reviewed, indicating at least that many companies/app developers are putting their apps on this service. With the growing reviews, its expected that the amount of apps will grow overtime.